

Closing the Care Loop — System Architecture v3

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Problem: Accessibility and quality of public healthcare services in rural and underserved areas.

Core thesis (revised): Rural care doesn't fail from missing services or missing apps — it fails because no system tracks whether a cross-tier *promise* (a referral, a consult, a follow-up, a stock claim) was actually kept. This platform is not another clinical delivery app. It is a **promise-tracking layer**: it records obligations created between tiers, verifies whether they were discharged using evidence each tier already generates, and escalates the misses to a named accountable person.

Stack: Flutter field app (ASHA-facing only) · Node.js API · lightweight FHIR-shaped store on PostgreSQL · React district dashboards · DLT-compliant SMS (staff/officer alerts only, never patient-facing) · store-and-forward handoff to eSanjeevani for teleconsult.

1 · Why the previous framing didn't hold

Two assumptions in the original design turned out to be false for the actual population, and both are corrected here rather than patched over:

"CHC will actively accept an incoming referral." They won't, and there's no reason they should — the patient gets treated on arrival regardless of whether anyone tapped a button first. An active-acceptance step measures app engagement, not care delivered. It either goes untapped (false escalation, alert fatigue) or gets rubber-stamped (meaningless data).

"The patient is reachable — has a phone, a stable address." For this population, mostly false. Any mechanism resting on SMS/IVR/OTP to the *patient* (not the worker) fails at the exact population the problem statement is about.

Both fixes point the same direction: **stop trying to track the patient digitally. Track the promise, and verify it using evidence the system already has — facility registration events and ASHA's own physical presence in the household — instead of inventing new channels to the patient.**

2 · The one mechanism the whole system is

Every component reduces to a single repeated shape, applied to four different promise types:



PROMISE MADE → DEADLINE SET → EVIDENCE OF KEEPING (or its absence) → ESCALATION ON MISS

Promise type	Who makes it	Evidence it was kept	Who's paged on miss
Referral	ASHA/ANM refers patient to CHC	Facility registration match (patient physically arrived)	Referring ASHA → Block MO → District nodal
Consult	Worker requests specialist review	eSanjeevani (or store-and-forward) returns a structured response within SLA	Requesting facility
Follow-up	Patient enrolled in a protocol (ANC/IMNCI/NCD)	ASHA attests outcome on next household visit	Supervisor (repeated misses only)
Stock accuracy	Facility reports drug X in stock	ASHA/patient encounters it missing on arrival	Block MO (stock-out logged, referral routing updated)

Nothing in this system performs triage, teleconsultation, or reminders better than the incumbents that already do those jobs (eSanjeevani, Kilkari, DVDMS). It sits *across* them, tracking whether the handoffs between them actually complete. This is the honest version of the "accountability layer" claim — every remaining component below has to justify itself against this table, or it's cut.

3 · Existing systems: the gap this attacks (condensed)

System	Covers	Documented gap
eSanjeevani	Doctor-to-doctor and OPD teleconsult, 43+ cr consultations	No screening/triage structure, no feedback loop to referring worker, "ends as an isolated single-step event" (PMC11422547)
ABDM	National health-record rails, registries, consent	Adoption ≠ integration — only 545/3,041 sandbox integrators have completed certification (NHA, Aug 2026)
Kilkari / Mobile Academy	Stage-driven IVR education to mothers with phones	One-way; not tied to actual encounters or referrals; assumes patient phone access , which this design no longer does either for its own comms
Khushi Baby CHIP	Offline CHW data platform, 85k CHWs, 60M+ tracked	Nearest real competitor. Strength is program-wise data collection; no documented cross-tier referral-SLA + escalation mechanism as a first-class object
ANMOL / RCH	ANM-specific reporting into RCH portal	No referral tracking, ASHAs remain on paper
DVDMS / e-Aushadhi	Warehouse-to-facility drug logistics	Reporting gaps, no last-mile visibility for the worker deciding whether to send a patient
Paper + phone calls	The actual daily incumbent	Zero infra cost, universally understood — loses on searchability, accountability, aggregation

The empty quadrant is unchanged from the earlier analysis: **no incumbent owns the seam between tiers**, because ownership requires both ends of a handoff inside one system, which every vertical silo structurally prevents.

4 · Scope: what is built vs. what is a named interface

This is the change that matters most for a hackathon build. The v2 document specified ten integrated subsystems. Most of that build risk bought no differentiation — it duplicated eSanjeevani, Kilkari, and DVDMS rather than tracking them. Scope is cut to what the promise-tracking layer actually requires.

Component	Status	Why
Referral state machine (SLA clock, arrival matching, escalation ladder)	Build — core	This <i>is</i> the product
ASHA field app: red-flag checklist → route decision → referral creation	Build — minimal	Just enough to create the obligation; not a full clinical decision-support suite
Facility-side registration code/QR matching	Build — core	This is what replaced "accept" — see §5
District dashboard (completion rate, unacknowledged escalations, stock-out flags)	Build — core	Where the promise-tracking pattern becomes visible and defensible live
ASHA follow-up task list (household visits due)	Build — minimal	Reuses existing worker workflow, doesn't invent a channel
eSanjeevani handoff	Stub / named interface	Request sent, structured response received, SLA tracked — do not rebuild their video pipeline
ABDM HIP adapter	Stub / named interface	Mock credentials, certification path documented, not attempted live
Stock accuracy	Single event type, not an inventory app	"Reported vs. found" mismatch logged at point of referral, not a warehouse system
Patient-facing SMS/IVR	Removed entirely	See §1 — the population this targets is largely unreachable this way
WebRTC video teleconsult	Removed from v1	eSanjeevani already owns this; rebuilding it duplicates, doesn't track

5 - Referral flow (revised)

5.1 Why "accepted" is gone for most referrals

An active accept step only carries real signal when the receiving side is committing a scarce resource in advance (a specialist slot, an OT booking, blood arranged). For the dominant case in this problem — a red-flag emergency referral where the patient is simply traveling — there is no meaningful gap between "referred" and "should be seen" for the CHC to schedule into, and no incentive for staff to interact with a queue before the patient shows up. Making that tap mandatory produces either alert fatigue (never tapped, everything reads as unacknowledged) or meaningless data (bulk-tapped, disconnected from actual arrival).

5.2 Revised state machine



Emergency / routine referral (majority case):

initiated —► arrived (passive, facility registration match) —► completed

└─ no arrival within SLA —► lapsed —► ESCALATED (ack-required)

Coordination-only referral (specialist slot, OT, blood arranged — minority case):

initiated —► accepted (active, resource commitment) —► arrived —► completed

initiated — ASHA/ANM creates the referral in the field app; a short code / QR is generated for the patient (works with SMS/paper fallback too — no smartphone or app dependency on the patient or CHC side).

arrived — captured passively: CHC front desk enters/scans the referral code as part of **normal OPD registration**, something already happening for every walk-in. No new workflow, no separate "accept" UI, no incentive problem.

completed — fires when the encounter is closed out at the facility; already-occurring clinical action, referral ID just gets attached to it.

lapsed — no **arrived** event within the SLA window. This is the real failure signal — the patient never made it, or made it and wasn't matched — and the correct trigger for escalation.

SLA defaults: red-flag 24h · urgent 48h · routine 7d, configurable per district.

5.3 Fallback when facility registration doesn't catch it

Some CHCs won't have registration discipline good enough to match codes reliably, or the patient never gets a slip at all. Second-tier evidence: **ASHA-attested outcome**, captured on her next routine household visit ("did you go, what happened") and entered next time she's online. This is lower-fidelity than a real-time registration match, but it is strictly higher-fidelity than the current baseline — a verbal referral with no record at all. This fallback is also how **follow-up adherence** is captured system-wide (§5.4) — it is not a separate mechanism, it's the same evidence path reused.

5.4 Follow-up (revised — no patient-facing reminders)

The v2 design generated IVR reminders to the patient. That's now removed — it's Kilkari's exact blind spot (comms tied to registry stage, not reachable patients) and this population is disproportionately unreachable that way. Follow-up instead generates a **task for the ASHA** — "visit household Y, check ANC status" — surfaced during her already-scheduled rounds, not a new channel. Outcome is entered by her, same as §5.3.

6 · Identity: what replaces phone/address matching

Dedup and patient identity do not rest on a phone number or a stable address — both are frequently absent for this population. Instead, identity is: **known to ASHA X, in household Y, within her existing catchment**. This isn't a new technical problem to solve — it's the relationship the CHW model already runs on informally.

The system's job is to make that relationship a queryable, shareable record instead of something that lives only in one worker's memory and is lost on transfer.

`[local_id]` (UUID) is the canonical identifier. `[abha_ref]` is nullable and populated opportunistically — not designed around, since OTP-based ABHA linking largely fails for a population without a personal phone. Coarse dedup (name + fuzzy-DOB + village) exists as a safety net for cross-worker matching, not as the primary identity mechanism.

Named risk, not hidden: this makes ASHA-continuity the single load-bearing assumption of the whole design. ASHA transfer or turnover can lose household context; catchment boundaries don't always match where people actually seek care; an ASHA may not physically reach a household before an SLA window closes. This is stated as an explicit trade-off, not solved — solving it is out of scope for this system.

7 · Component justification (only what survives §4)

Referral state machine separate from the field app: SLA timers and escalation are server-side, single-writer — devices enqueue intents, the server serializes. No distributed locking needed, and the state survives even if a device is lost.

Facility registration matching over an "accept" screen: reuses a workflow CHC staff already do for every patient, at zero marginal effort, and produces evidence tied to what actually happened rather than what someone remembered to tap.

ASHA task list over patient-facing comms: the only actor reliably reaching this population without a phone number or fixed address is the frontline worker physically visiting. Everything patient-facing collapses into tasks for her, not messages to them.

eSanjeevani/ABDM as named interfaces, not rebuilt subsystems: the system's job is to know whether a promise made to those systems was kept, not to duplicate what they already do well (or, per PMC11422547, poorly — either way, that's their problem to fix, not this system's to re-solve).

8 · Security, privacy, abuse controls

Threat	Control
PHI on cheap shared Android devices	SQLCipher AES-256 at rest · device-bound PIN + remote wipe · only the ASHA's active caseload syncs to device
Transport interception	TLS 1.3 + certificate pinning
Incentive gaming (ghost referrals/arrivals)	GPS+timestamp plausibility checks on creation · supervisor spot-audit sampling · anomalies flagged, not auto-blocked
DPDP Act 2023 posture	Consent artifact per interaction purpose · guardian consent for minors · purpose limitation via role scoping · append-only signed audit trail
API abuse	Per-device rate limits; emergency-class traffic exempted but flagged

No patient SMS/IVR channel exists in this design, which removes an entire prior privacy surface (§1 of v2) by construction rather than by control.

9 • Core data model (trimmed to what's built)

Entity	Key fields	Notes
Patient	<code>local_id</code> (UUID, always present) · <code>abha_ref</code> (nullable) · name/approx-age/sex · household + ASHA catchment link (not address)	Identity strategy per §6
Referral	<code>from</code> / <code>to</code> facility · priority (routine/urgent/red-flag) · <code>status</code> ∈ initiated → arrived → completed lapsed (+ optional <code>accepted</code> for coordination cases) · <code>sla_deadline</code> · escalation ladder with ack timestamps	The core object
Encounter	type (triage/consult/in-person) · facility · worker · timestamp	Append-only
FollowUpTask	due date · assigned ASHA · outcome (adherent/missed/rescheduled) entered on next visit	No patient channel field — removed
StockEvent	facility · drug code · <code>reported</code> vs <code>found</code> mismatch · logged at point of referral	Single event type, not an inventory system
ConsentArtifact / AuditEvent	purpose, scope, timestamps, signatures	Unchanged from v2

10 · Gaps found and fixes applied (running log, including this round)

Gap	Why it broke	Fix
CHC has no incentive to actively "accept" a referral before the patient arrives	Measures app interaction, not care delivered; produces false escalations or meaningless rubber-stamped data	Passive arrival capture via existing OPD registration (§5.2)
Patient often has no phone or stable address	SMS/IVR/OTP-based mechanisms fail exactly for the target population	Removed patient-facing comms entirely; ASHA becomes the evidence and follow-up channel (§5.3, §5.4)
Ten integrated subsystems is not a hackathon build	Build risk concentrated in components that don't differentiate the system	Scoped to core state machine + minimal field capture + named interfaces for teleconsult/ABDM (§4)
ABDM sandbox certification is a months-long process	Only 545/3,041 integrators have completed it	Local-ID-first, ABHA linkage opportunistic, adapter is a documented interface not a dependency
Identity without phone/address	Name+DOB+village matching alone is low-precision	ASHA-catchment relationship as primary identity signal, dedup as safety net only

Trade-offs consciously accepted

ASHA-attested confirmation is lower-fidelity than real-time digital tracking — accepted because it is still strictly better than the current baseline of no record at all.

The entire design now depends on ASHA continuity and catchment accuracy — a real risk (transfer, turnover, boundary mismatch), named here rather than solved.

Coordination-referral accepted step reintroduces the original incentive question on a smaller surface (specialist slots, OT bookings) — accepted because there a real resource commitment justifies the extra tap.

11 · USP statement

Unlike eSanjeevani (isolated consultation events), Kilkari (one-way IVR education requiring a patient phone), or Khushi Baby (program-wise data collection per state deployment), this system tracks whether **cross-tier promises — referral, consult, follow-up, stock accuracy — are actually kept**, using evidence each tier already generates (facility registration, worker household visits) rather than inventing new channels to a population that is frequently unreachable by phone. It does not compete with any incumbent's clinical delivery; it is the layer that notices when the handoff between them fails, and tells a named person.

12 · Open questions requiring a decision before build

Pilot district — determines which facility registration workflow (paper register vs. existing EMR) the arrival-matching mechanism has to integrate with first.

Demo teleconsult — stub the eSanjeevani handoff with a mocked response, or get a real (even informal) doctor-side contact for the demo round?

Minimum device floor for the ASHA app (Android Go / 1GB RAM) — constrains build choices now, not later.

How much of the coordination-referral (`accepted`) path is worth demoing vs. leaving as a documented variant, given it's the minority case?

Language set for v1 field app — pick 2 rather than promise many.

Summary

Problem: referrals, consults, follow-ups, and stock claims cross tiers in rural care and nobody tracks whether they were kept — every incumbent is a silo that owns one side of the handoff.

Architecture: a server-side promise-tracking state machine (referral/consult/follow-up/stock, each `promise → deadline → evidence → escalation`), fed by passive facility-registration matching and ASHA household-visit attestation rather than patient-facing digital channels, with eSanjeevani and ABDM integrated as named interfaces rather than rebuilt.

USP: the only system tracking whether cross-tier care obligations were kept, using evidence that already exists at each tier, for a population that a phone-based or app-based patient channel cannot reach.